

A Tabu Search Scheduling Algorithm For MIMO CDMA Systems

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Abstract—In multiuser multiple input multiple output (MIMO) systems, it is optimal to serve multiple users at the same time in order to achieve high data rates. However, the use of a transmit beamforming technique requires a well designed user selection scheme to obtain good performances. The optimal scheduling solution can only be obtained through a highly computationally complex exhaustive search. In addition, when employing a multiple access scheme, such as the code division (CDMA), the complexity of an optimal user selection becomes higher even for moderate number of users and antennas. In this context, this paper proposes a heuristic scheduling algorithm based on a tabu search approach for MIMO CDMA systems using ZFBF as a transmit technique. We use a graph theoretical approach to model the system as a weighted undirected graph. The problem of user selection is then formulated as a graph coloring problem. Numerical results show that the proposed algorithm outperforms the greedy scheduling scheme and achieves performances, in terms of system sum rate, very close to those of the highly complex optimal solution.

I. INTRODUCTION

A MIMO multiuser system can be viewed as a wireless system where a multi-antenna base station (BS) communicates with several mobile users. These users, due especially to their small size, are constrained to be equipped with a limited number of antennas. Hence, in the case of a multiuser broadcast channel (BC), if the BS serves only one user at a time, the resulting system rate will be penalized by the limited number of antennas at the receiving point. Therefore, in order to achieve high system performances, the BS has to serve multiple users. Furthermore, in addition to the MIMO gain, the BS can take benefit from a kind of selection diversity called multiuser diversity. Nevertheless, designing the best scheduling strategy that extracts the two mentioned gains remains a very challenging research topic.

When the BS knows perfectly the channel coefficients between its antennas and those of the mobile users, zero forcing beamforming (ZFBF) is known to be a suboptimal transmit technique that solves the tradeoff between feasibility and performance [1]. Also, [1] demonstrates that ZFBF achieves the same sum rate as, the optimal but very complex, dirty paper coding (DPC) [2] as the number of users goes to infinity. However, the performances of this transmit scheme depends extremely on the choice of the set of users to be served.

The optimal set of users when employing ZFBF in MIMO BC can be obtained by performing an exhaustive search among the possible combinations of users. Since this optimal solution has a very high complexity, design of low complex

scheduling schemes is of great interest. Effectively, several user selection algorithms were proposed in the literature. The authors in [3] present and analyze the performances of two reduced complexity scheduling algorithms designed for multiuser MIMO systems employing DPC or ZFBF.

On the other hand, multiuser MIMO systems using code division, also known as MIMO-CDMA systems, received a great research interest over the last few years. However, most research works on this area has focused on the design of detection receivers to increase the number of supported users on the system [4]. This type of architectures requires complex processing at the receivers' side instead of using a precoding scheme at the transmitter where more intelligence can be added more easily. Therefore, in this paper we are interested in designing an efficient scheduling algorithm to be implemented at the BS of a MIMO-CDMA system. Due to the code dimension, the complexity of finding the optimal set of users to be served becomes more complex. Hence, we propose a suboptimal tabu search metaheuristic algorithm (TSA) to address the problem of user selection. As metaheuristic algorithms, TSAs are known for finding a very good solution to optimization problems with relatively low computational complexity. In the context of user selection for ZFBF, a metaheuristic algorithm (a genetic one) was first used in [5] to propose a scheduling solution for MIMO BC using ZFBF. Recently, [6] has extended the genetic algorithm in [5] to the case of DPC.

In this paper, we adopt a graph theoretical approach where the MIMO CDMA studied system is modeled as a weighted graph. This graph representation allows us to define a graph coloring problem equivalent to the user selection problem. The formulated problem is known as the maximum weight k -colorable subgraph problem. Hence, since tabu search algorithms are well known to provide good results for graph coloring problems, especially for dense graphs [7] which is our case, we design a tabu search colouring to find a near-optimal solution to the scheduling problem.

The paper is organized as follows. The system model is formulated in Section II. Section III presents a graph representation of the system, formulates the graph colouring problem and details the proposed scheduler. Simulation results are presented in Section IV, and Section V concludes the paper.

II. SYSTEM MODEL

We consider a wireless transmission system consisting of a multiple antenna BS which tries to serve a pool of K

users. We assume that the BS is equipped with M transmit antennas, whereas each user has only single antenna. The channel coefficients between the BS transmit antennas and the users' antennas are modeled as independent identically distributed (i.i.d.) complex Gaussian variables with zero mean and unit variance. We consider that time is divided into slots. The channel coefficients are assumed to be constant during the entire slot but may change from one slot to another.

We assume that data is always available, at the BS, ready to be transmitted to the scheduled users. Therefore, the scheduling decision is isolated from the effect of data arrival. Let $\mathbf{h}_k$ denote the channel coefficients $M \times 1$ vector between the BS transmit antennas and user k antenna. It is assumed that the BS perfectly knows the channel vectors for all users.

Since we assume the use of CDMA as a multiple access technique, the BS makes use of N spreading code. At each time slot, the BS must select a given number of users to be served. The scheduled users are then disposed in at most N (same as the number of available codes) distinct sets; denoted by ζ_n , ($n = 1, \dots, N$). Each scheduled set of users is then served using the same code denoted by the $1 \times C$ vector $\mathbf{c}_n$ where C represents the processing gain. The used spreading codes are a combination of orthogonal codes and pseudo-random noise (PN) codes. Note that we use this combination to improve the correlation properties of the signals.

We denote by $\mathbf{S}_n$ the $M \times C$ matrix of the chip-level transmitted signals from the BS to the users belonging to the set ζ_n . In fact, each user's signal is first spread using the corresponding spreading code. The resulting signal is then multiplied by a precoding vector before transmission.

$$\mathbf{S}_n = \sum_{k \in \zeta_n} \sqrt{P_k} \mathbf{w}_k \mathbf{c}_n x_k \quad (1)$$

where P_k is the power allocated to user k , $\mathbf{w}_k$ is the $M \times 1$ precoding vector and x_k is the transmitted data symbol for the k^{th} user.

The BS has a limited amount of power to allocate to the scheduled users at each timeslot denoted by P . Also, P_n denotes the portion of the total power P that the BS allocates to the users belonging to the set ζ_n . Then, we have

$$P = \sum_{n=1}^N P_n = \sum_{n=1}^N \sum_{k \in \zeta_n} P_k. \quad (2)$$

Therefore, the chip-level sampled signal $1 \times C$ vector $\mathbf{y}_k$, for this user is given by:

$$\begin{aligned} \mathbf{y}_k = & \mathbf{h}_k^T \sqrt{P_k} \mathbf{w}_k \mathbf{c}_n x_k + \sum_{j \in \zeta_n, j \neq k} \mathbf{h}_k^T \sqrt{P_j} \mathbf{w}_j \mathbf{c}_n x_j \\ & + \sum_{l=1, l \neq n} \sum_{j \in \zeta_l} \mathbf{h}_k^T \sqrt{P_j} \mathbf{w}_j \mathbf{c}_l x_j + \mathbf{z}_k \end{aligned} \quad (3)$$

where $\mathbf{z}_k$ is the $1 \times C$ vector representing the i.i.d. additive white complex Gaussian noise with zero mean and unit variance.

In order to totally eliminate the interference between the users using the same spreading code, we assume that the BS

makes use of ZFBF as a precoding scheme. Hence, the interference among the users sharing the same code, represented by the second term in (3), is eliminated through ZFBF. Whereas, the interference produced by the users in the other sets (the third term in (3)) is limited since we use CDMA. Note that ZFBF limits the number of users that can be scheduled in a set to the number of antennas M , i.e. $Card(\zeta_n) < M$ where $Card(\cdot)$ denotes the number of elements on a set.

For clarity, denote by $\mathbf{h}_{\zeta_n}^{(k)}$ the channel coefficients vector for user k belonging to the set ζ_n . $\mathbf{H}_{\zeta_n}$ represents the $M \times Card(\zeta_n)$ matrix formed by the channel vectors of the users in ζ_n . Therefore, the corresponding beamforming matrix formed by the precoding vectors and denoted by $\mathbf{W}_{\zeta_n}$ is obtained by performing a pseudo-inversion of $\mathbf{H}_{\zeta_n}$. Hence the k^{th} column of $\mathbf{W}_{\zeta_n}$ is given by

$$\mathbf{w}_{\zeta_n}^{(k)} = \frac{\mathbf{h}_{\zeta_n}^{\dagger(k)}}{\sqrt{\|\mathbf{h}_{\zeta_n}^{\dagger(k)}\|_F^2}} \quad (4)$$

where $\mathbf{h}_{\zeta_n}^{\dagger(k)}$ is the k^{th} column of the pseudo inverse of $\mathbf{H}_{\zeta_n}$.

The received signal at the k^{th} user after being processed by a matched filter can be given as

$$\begin{aligned} \mathbf{y}_k \cdot \mathbf{c}_n^T = & \sqrt{P_k} x_k \\ & + \sum_{l=1, l \neq n} \sum_{j \in \zeta_l} \sqrt{P_j} \mathbf{h}_k^T \mathbf{w}_j x_j \sqrt{\alpha} \rho \\ & + \mathbf{z}_k \mathbf{c}_n^T \end{aligned} \quad (5)$$

where ρ is the cross correlation among the different PN scrambling codes and α represents the degree of imperfection of the orthogonality among the used orthogonal codes also called the orthogonality factor.

The maximum achievable sum rate for each scheduled user when using ZFBF is given by

$$R_{\zeta_n}^{(k)} = \log_2 \left(1 + \frac{P_k}{1 + \alpha \rho^2 \cdot \sum_{l=1, l \neq n} \sum_{j \in \zeta_l} P_j |\mathbf{h}_k^T \mathbf{w}_j|^2} \right) \quad (6)$$

Hence the sum rate is maximized when scheduling, in the same set, users having near orthogonal channel coefficients. Also, note that as the value of the orthogonality factor decreases, the sum rate increases. The maximum achievable sum rate for the overall scheduled users is given by

$$R_T = \max_{P_k: \sum_{n=1}^N \sum_{k \in \zeta_n} P_k \leq P} \sum_{n=1}^N \sum_{k \in \zeta_n} R_{\zeta_n}^{(k)} \quad (7)$$

The optimal power allocation $P_{k,opt}$ for the k^{th} user can be found through a continuous optimization as the one proposed in [8]. Such a solution can add more complexity to the scheduling algorithm, hence we use in this paper a simple water filling strategy to find a near optimal power distribution. The water filling equation is given by:

$$P_{k,opt} = (\mu / \|\mathbf{w}_{\zeta_n}^{(k)}\|^2 - 1)^+ \quad (8)$$

where μ is the solution of the equation given as follows:

$$\sum_{n=1}^N \sum_{k \in \zeta_n} (\mu - \|\mathbf{w}_{\zeta_n}^{(k)}\|^2) = P \quad (9)$$

III. THE ZFBF MIMO CDMA TABU SEARCH ALGORITHM (TABU-ZFBF-MC)

In this section, we first formulate the scheduling problem as a graph colouring problem. Then, we present a low complexity metaheuristic algorithm based on the well-known tabu search approach.

A. Graph Representation

We first present a graph representation of the MIMO CDMA system. The resulting graph $G = (V, E)$, where V is the set of vertices and E is the set of edges, is called the system graph and it can be obtained as follows [9]: Each user k in the system is represented by a vertex v_k and we assign to each vertex a non-negative weight given by its corresponding channel gain $\|\mathbf{h}_k\|^2$. Furthermore, there is an edge (v_i, v_j) between vertices v_i and v_j if and only if

$$e_{ij} = \frac{|\mathbf{h}_i \mathbf{h}_j|}{\|\mathbf{h}_i\| \|\mathbf{h}_j\|} > \epsilon, \quad (10)$$

i.e. their channels are not ϵ -orthogonal where ϵ is a constant denoting the orthogonality threshold.

Note that the value of the parameter ϵ has a direct impact on the performances of the system. In fact, since the selected users in each set are not perfectly orthogonal, the scheduling decision will not be always optimal introducing some rate loss influenced by the chosen value of ϵ . Anyhow, we use a near-optimal value of ϵ found by simulations.

B. Graph Problem Formulation

Let us consider a graph $G = (V, E)$ similar to the one defined earlier where every vertex v in V is assigned a non negative weight γ_v . We consider also an integer number $k > 0$. We define the maximum weight k -colorable subgraph problem as finding a subgraph $G' = V' \subset V, E'$ of the graph G , such that there exists a vertex coloring $C_{V'}$ of G' with k colors that maximizes the value $\sum_{v \in V'} \gamma_v$ among all the possible subgraphs. Such a problem can be typically given by the following mathematical formulation:

$$\begin{aligned} &\text{Find } V' \subseteq V \\ &\text{such that } \exists C_{V'} \text{ with } k \text{ colors and } \max_{V'} \sum_{i \in V'} \gamma_i \end{aligned}$$

It has been shown in [10] that this problem is NP-hard.

The scheduling algorithm have the mission of choosing the scheduled users disposed in a given number of sets. Therefore, finding the best sets of users for a MIMO CDMA system using ZFBF can be seen as the problem of finding a solution to the maximum weight k -colorable subgraph problem having as

input the system graph and the number of available spreading codes which corresponds to the number of colors to use.

C. A Tabu Search Algorithm (Tabu-ZFBF-MC)

[7] was the first work that adapts the tabu search technique to the problem of graph colouring (TABUCOL). The proposed algorithm gives good solutions and outperforms several heuristics especially in the case of dense graphs. In this paper, we develop a solution to the maximum weight k -colorable subgraph problem based on a tabu search approach.

In order to adapt tabu search metaheuristics to solve a given problem, the developed algorithm has to:

- construct an initial solution,
- define moves that determine the neighbors of a solution,
- decide of the content and size of tabu list, and finally,
- design diversification mechanisms.

The tabu search ends when one of the following two conditions occurs.

- The maximal number of allowed iterations has been reached.
- The maximal number of iterations, where the best solution is not enhanced successively, has been reached.

In the following, we detail how we adapt the tabu search metaheuristic steps in order to solve the problem of user scheduling in ZFBF based MIMO CDMA systems.

1) *Initial Solution*: Given a system graph $G = (V, E)$ and a number of colors k (which corresponds also to the number of available spreading codes), the first step of a tabu search is to find an initial solution to the algorithm. Remember that the initial solution must be feasible, i.e. the initial k -colouring must be legal respecting the constraints of graph colouring. We propose two ways to construct the initial k -colouring: random colouring and greedy colouring. The random colouring assign a different color to a user picked randomly while keeping the colouring legal. Whereas, the greedy colouring runs the greedy algorithm proposed in [9] to find an initial solution. Though the greedy colouring gives a better initial solution, we use the random colouring due to its lower complexity.

2) *Move Definition*: The definition of the neighborhood $Ne(s)$ of the current solution s is a crucial step since it has an impact on the quality of the final solution. Note that s represents the current k -colouring of the graph. In this paper, we generate a neighbour s' of the current solution s as follows: we choose a non-coloured vertex v_i . Then we compute the sum weights for each possible colouring of v_i . Colouring v_i with a given color assumes decolouring its adjacent vertices having the same color. Hence, each move is represented by a pair (v_i, c_n) with $n = 1, \dots, N$.

Having generated all the possible moves, excluding obviously tabu moves, we pick up the best one (the one improving the sum weights) and we move to it.

3) *Tabu List*: The tabu list is obtained as follows. Whenever an initially non-coloured node v_i takes a color c_n to get the new solution, the pair (v_i, c_n) is added to the tabu list, i.e. if the v_i loses its color then it can not be coloured by c_n

Algorithm 1: The ZFBF MIMO-CDMA Tabu Search

Input:

$Itermax$ = maximum number of iterations
 $G = (V, E)$
 k = number of colors (codes)
 $|T|$ = size of the tabu list

Initialization:

Generate a random legal k -colouring
Initialize the tabu list T
 $Iter \leftarrow 0$
 $Bad_move \leftarrow 0$
 $Change \leftarrow true$

while $Iter < Itermax$ **do**

 for each non coloured vertex v_i **do**

Generate possible non taboo moves

Update the coloring:

Step1: Choose the best move: (v_i, c_n) the one that improves the sum weights (assuming always a legal k -coloring)

 Step2: Colour v_i with color c_n (Put user v_i in the set c_n)

 Step3: Decolour the adjacent vertices of v_i already coloured by c_n

 Step4: Update T : introduce (v_i, c_n) and remove the oldest tabu move

 if the colouring is updated **then**

 $Change \leftarrow true$

 $Best_Colouring \leftarrow Current_Colouring$

 end

 end

 if $Change = false$ **then**

Take the best move in the current iteration (in spite if it doesn't improve the sum weight)

 $Bad_move \leftarrow Bad_move + 1$

 end

 if $Bad_move = Allowed_bad_moves$ **then**

Diversification

end

 $Iter \leftarrow Iter + 1$
end

Output: The sets of users to be served are given by the sets in $Best_Colouring$

for a given number of iterations. Note that each time a new pair (vertex, color) is added to the tabu list, the oldest pair is removed from the list.

4) *Diversification*: Diversification is a mechanism that permits to improve a tabu search method. In the proposed algorithm, we allow sometimes a move that does not improve the current solution. Such a move is considered as a bad move. When the number of bad moves reaches a certain fixed threshold, we perform diversification. In fact, in order to get out from a local maximum, a random k -colouring, like the one that constructs the initial random solution, is performed.

The complete formulation of the tabu search algorithm that solves the user selection problem is given by Algorithm 1.

IV. SIMULATION RESULTS

In this section, we analyze the performance of the proposed tabu search algorithm in terms of the maximum achievable throughput. The simulations are performed for a relatively simple system with four spreading codes in order to reduce the simulation complexity, especially for the exhaustive search.

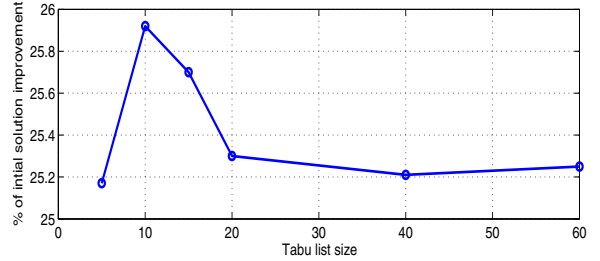
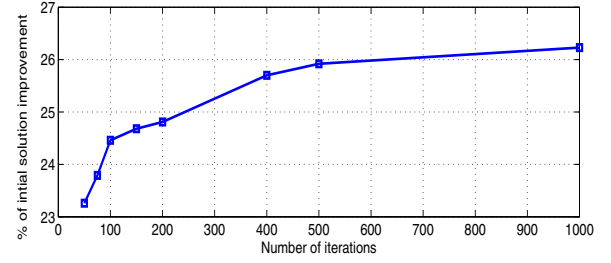


Fig. 1. Impact of the tabu search parameters on the quality of the solution

The performances of the proposed algorithm are compared to those of the greedy algorithm, presented in our previous work [9], and to the optimal performances obtained by the very complex exhaustive search. We use a near optimal value for the orthogonality threshold ϵ obtained through simulations (i.e. $\epsilon = 0.5$ when $M = 2$ and $\epsilon = 0.375$ when $M = 4$). The value of the cross correlation between PN codes is taken equal to $1/\sqrt{C}$ where C is the processing gain. Also, the orthogonality factor α is set to 0.1.

Remember that two parameters have to be adjusted, before implementing the proposed algorithm, as they have a direct impact on the quality of the final solution. These two parameters are the size of the tabu list $|T|$ and the maximum number of allowed iterations $Itermax$. Therefore, it is important to quantify their impact and then adjust their values accordingly. Fig. 1 shows the percentage of improvement of the initial solution when varying the two parameters. the number of users is fixed to 20 and the BS is equipped with four antennas. We observe that the maximum percentage of improvement is obtained when setting the tabu list size to 10. In fact, a larger size reduces the number of neighbors to visit at each iteration whereas a smaller size results on cycles which not improve the final solution. On the other hand, we notice that as the number of iterations increases, the performance of the proposed scheduler becomes higher. However, this improvement becomes negligible for larger values of $Itermax$. Hence, choosing a large $Itermax$ may introduce complexity without a noticeable performance improvement. Hence, we use an $Itermax$ equal approximately to 20 times the number of users.

Fig. 2 presents a comparison between the proposed tabu search algorithm and the greedy algorithm in terms of sum rate when varying the number of users to serve. We consider a BS equipped with two transmit antennas (lower curves) or

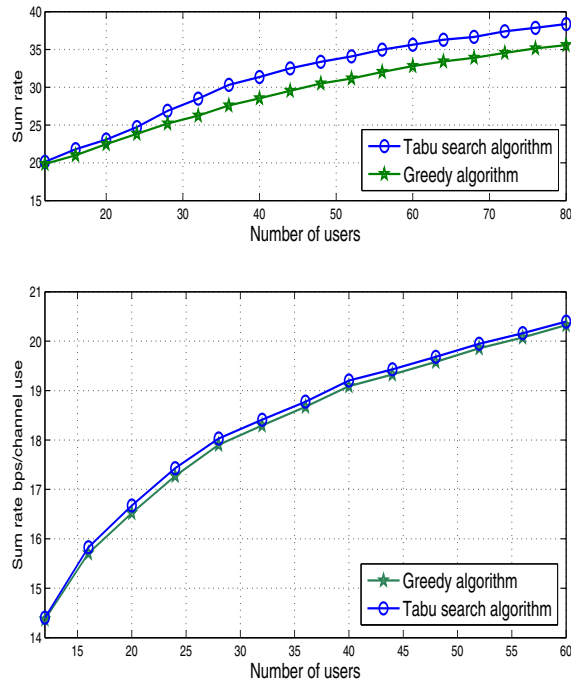


Fig. 2. Sum rate Vs number of users to be served by a 2 Tx or 4 Tx BS

four transmit antennas (higher curves). It is observed that the proposed algorithm always outperforms the greedy algorithm. The gap between the performances of the two algorithms becomes more significant as we add more users to the system. Indeed, as the size of the system graph becomes larger, the exploration performed by the tabu search approach becomes more interesting since it makes more moves in order to improve the quality of the final solution. Therefore, the gap between the performances of the two scheduling schemes can be as high as 10% (in the case of 4Tx BS). Furthermore, we notice that the tabu search algorithm like the greedy algorithm takes advantage from the multiuser diversity since its performances are still increasing as we have more users.

Finally, We plot in Fig 3 the sum rate achieved by the proposed algorithm in order to compare it with the one obtained through an optimal user selection. Remember that this optimal solution is reached by performing a highly complex exhaustive search among all the possible users' combinations. Due to the high complexity of this scheme, the number of users to be served is limited to 16. The figure illustrates the near optimality of the proposed algorithm since it achieves sum rates very close to those of the exhaustive search. We also notice that the gap between the tabu search algorithm performances and the optimal solution remains almost the same for different number of users.

V. CONCLUSION

We have presented a heuristic scheduling algorithm to solve the user selection problem in MIMO CDMA systems using

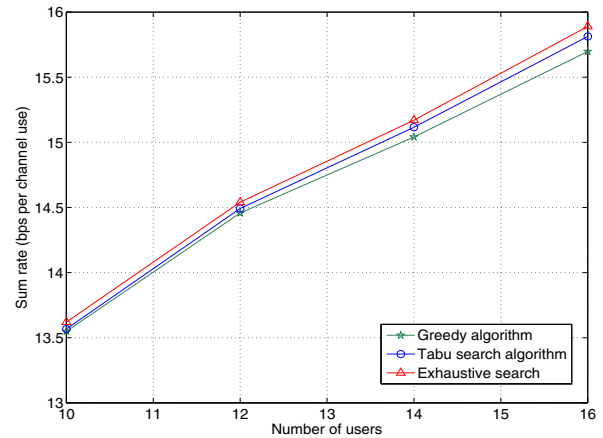


Fig. 3. Sum rate Vs number of users served by a 2 Tx BS

ZFBF as a transmit technique. In particular, the proposed algorithm adopts a tabu search approach in order to find a near optimal solution to the scheduling problem with a very low computational complexity since the optimal solution is known to be very complex and thus infeasible. Before applying the proposed algorithm, we use a graph theoretical approach in order to formulate the user selection problem as the maximum weight k -colorable subgraph problem. Through simulations, we have shown that the proposed tabu search algorithm achieves higher sum rates than the greedy algorithm. Also, the simulations show that the presented algorithm achieves performances very close to those of the optimal but very complex exhaustive search.

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